Complete list of publications of Shuang Song

Shuang Song

October 12, 2026

A) Publications with peer review process

under review

[1] Hou, Y., *Wang, S., **Song, S.**, Jiao, C., Wu, X., Roberts, P., and Fu, B. (under review). "Crop Optimization Improves Agricultural Performance but Intensifies Irrigation Water Mismatches in the Yellow River Basin". In: *Agricultural Water Management*.

[2] Hu, B., ***Song, S.**, Thomas, T., Wang, Y., Zhang, X., Liu, Y., Wang, S., Geng, L., and Roberts, P. (under review). "Psychological Differences in Implicit Individualism and Holistic Thinking between Herding and Irrigation-Farming Regions in China". In: *PNAS Nexus*.

[3] Li, Z., Pan, W., Li, X., Wang, Y., Wang, T., Victor, B., Zhang, H., Zhu, X., Zhang, Z., Niu, Y., **Song, S.**, Wang, S., and Bruijnzeel, A. (under review). "Permafrost Degradation Drives Emergent Streamflow Transitions across Northern High Latitudes". In: *Nature Geoscience*.

[4] *Qin, Y., Findley, D. M., **Song, S.**, Stevenson, J., Hamilton, R., Pagunsan, R., Barreto-Tesoro, G., Bankoff, G., and Roberts, P. (under review). "Colonial Deforestation Amplified Warming and Hydroclimatic Risks in the Philippine Archipelago". In: *Science Advance*.

[5] ***Song, S.**, Wang, S., Li, Z., Wu, X., Luo, Y., Hou, Y., Hu, B., He, Z., Bo, Q., Schaefer, G., Jiao, C., Vogl, M., Li, Y., Renn, J., Roberts, P., and Fu, B. (under review). "Water Quotas Improve Irrigation but Undermine Groundwater Sustainability". In: *Communication Sustainability*.

[6] Zhang, L., Hu, H., **Song, S.**, and *Cao, J. (under review). "Human-Water Interactions and the Divergent Strategies of Social Organization in the Late Neolithic Yangtze River Valley". In: *Journal of Flood Risk Management*.

submitted

[7] Chen, L., *Wu, X., Wang, S., **Song, S.**, Bennett, E., and Fu, B. (submitted). "Limited Representativeness and Effectiveness of Chinas National Sustainable Development Pilots". In: *Proceedings of the National Academy of Sciences*.

[8] Que, J., **Song, S.**, Niu, Y., Zhang, X., Roberts, P., and Xiao, L. (submitted). "From Breach to Bureau: Institutional Adaptation for Yellow River Flood Defence before 1855 in the Qing Dynasty, China". In: *Journal of Flood Risk Management*.

[9] Zhang, L., ***Song, S.**, Ma, T., and Roberts, P. (submitted). "An Agent-Based Model Reveals the Complex History of Delayed Agricultural Diffusion in Tropical South China". In: *Journal of Computer Applications in Archaeology*.

in press

[10] ***Song, S.**, Hu, B., Findley, D. M., Wu, X., Liu, W., Middeke-Conlin, R. W., Caetano, V. L., Yang, Y., Wang, S., Renn, J., and Roberts, P. (in press). "Archival and Palaeoenvironmental Documentation of Extreme Climate Events Reveals Generational Amnesia in Collective Memory". In: *Nature Communications*.

2026

[11] Hu, B., ***Song, S.**, Wang, Y., Zhang, X., Liu, Y., Wang, S., Geng, L., and Roberts, P. (2026). "Memory Sources Shape Flood Risk Perception and Agricultural Decision Among Farmers in a Drought-Oriented Irrigation District". In: *Journal of Flood Risk Management* 19.3, e70244. DOI: 10.1111/jfr3.70244.

[12] Li, Y., Liu, S., Wu, X., **Song, S.**, Zhao, Y., Li, Y., Wang, W., Fan, J., Tian, Z., Lin, J., Jin, Z., Wu, G., and Tran, L.-S. (2026). "Spatial Trade-Offs between Environmental Sustainability and Crop Production in China under SDG-Aligned Scenarios". In: *iScience* 29.8, p. 116802. DOI: 10.1016/j.isci.2026.116802.

[13] Sang, S., *Li, Y., Zong, S., Yu, L., Wang, S., Liu, Y., Wu, X., **Song, S.**, Zhao, W., Wang, X., and Fu, B. (2026). "CHANS-SD-YRB V1.0: A System Dynamics Model of the Coupled Human-Natural Systems for the Yellow River Basin". In: *Geoscientific Model Development* 19.5, pp. 2039–2058. DOI: 10.5194/gmd-19-2039-2026.

[14] ***Song, S.**, Zhang, L., Renn, J., and Roberts, P. (2026). "From Regime Shifts to Adaptive Cycles: Dynamic Understanding of Human–Water System". In: *Environmental Research: Water* 2.2. DOI: 10.1088/3033-4942/ae609a.

[15] Yu, L., Wang, S., **Song, S.**, Ye, Q., Wu, X., and Fu, B. (2026). "ISEHSD: A Feedback-Based, Integrated Social-Ecohydrological System Dynamic Model for Studying River Basin-Level Transformation". In: *Environmental Modelling & Software* 204, p. 107103. DOI: 10.1016/j.envsoft.2026.107103.

2025 [16] Hou, Y., Wang, S., **Song, S.**, Chen, P., and Wu, X. (2025). "Crop Irrigation Water Requirements Mismatch the Actual Water Allocation in the Anthropogenic-Regulated Yellow River Basin". In: *Journal of Hydrology: Regional Studies* 61, p. 102715. DOI: 10.1016/j.ejrh.2025.102715.

[17] Sang, S., Li, Y., Zong, S., Yu, L., Wang, S., Liu, Y., Wu, X., **Song, S.**, Wang, X., and Fu, B. (2025). "The Modeling Framework of the Coupled Human and Natural Systems in the Yellow River Basin". In: *Geography and Sustainability* 6.4, p. 100294. DOI: 10.1016/j.geosus.2025.100294.

[18] Wang, S., **Song, S.**, Liu, Y., Wu, X., Jiang, E., and Fu, B. (2025). "Scientifically Improving Integrated Water Governance for Resilient Flow of the Yellow River". In: *Bulletin of Chinese Academy of Sciences* 40.8, pp. 1347–1356. DOI: 10.3724/j.issn.1000-3045.20241224001.

[19] Wang, S., **Song, S.**, Zhang, H., Yu, L., Jiao, C., Li, C., Wu, X., Zhao, W., Best, J., Roberts, P., and Fu, B. (2025). "Anthropogenic Impacts on the Yellow River Basin". In: *Nature Reviews Earth & Environment*, pp. 1–16. DOI: 10.1038/s43017-025-00718-2.

[20] Wu, X., Yan, Z., Yang, H., Wang, S., Zhang, H., Shen, Y., **Song, S.**, Liu, Y., Guo, Y., Yang, D., and Fu, B. (2025). "Ecological Restoration in the Yellow River Basin Enhances Hydropower Potential". In: *Nature Communications* 16.1, p. 2566. DOI: 10.1038/s41467-025-57891-7.

2024 [21] Jiao, C., Wu, X., **Song, S.**, Wang, S., Xiang, B., and Fu, B. (2024). "River Stabilization Reshaped Human-Nature Interactions in the Lower Yellow River Floodplain". In: *Journal of Environmental Management* 371, p. 122957. DOI: 10.1016/j.jenvman.2024.122957.

[22] **Song, S.**, *Wang, S., Jiao, C., and Ibarra, E. J. M. (2024). "ABSESpy: An Agent-Based Modeling Framework for Social-Ecological Systems". In: *Journal of Open Source Software* 9.96, p. 6298. DOI: 10.21105/joss.06298.

[23] **Song, S.**, Wen, H., Wang, S., Wu, X., Cumming, G. S., and Fu, B. (2024). "Quantifying the Effects of Institutional Shifts on Water Governance in the Yellow River Basin: A Social-Ecological System Perspective". In: *Journal of Hydrology* 629, p. 130638. DOI: 10.1016/j.jhydrol.2024.130638.

2023 [24] Jiao, C., *Wang, S., **Song, S.**, and Fu, B. (2023). "Long-Term and Seasonal Variation of Open-Surface Water Bodies in the Yellow River Basin during 1990–2020". In: *Hydrological Processes* 37.3, e14846. DOI: 10.1002/hyp.14846.

[25] **Song, S.**, *Wang, S., Wu, X., Wei, Y., Cumming, G. S., Qin, Y., Wu, X., and Fu, B. (2023). "Identifying Regime Transitions for Water Governance in the Yellow River Basin, China". In: *Water Resources Research* 59.12, e2022WR033819. DOI: 10.1029/2022WR033819.

[26] Wu, X., *Fu, B., Wang, S., **Song, S.**, Lusseau, D., Liu, Y., Xu, Z., and Liu, J. (2023). "Bleak Prospects and Targeted Actions for Achieving the Sustainable Development Goals". In: *Science Bulletin* 68.22, pp. 2838–2848. DOI: 10.1016/j.scib.2023.09.010.

2022 [27] Chen, P., *Wang, S., **Song, S.**, Wang, Y., Wang, Y., Gao, D., and Li, Z. (2022). "Ecological Restoration Intensifies Evapotranspiration in the Kubuqi Desert". In: *Ecological Engineering* 175, p. 106504. DOI: 10.1016/j.ecoleng.2021.106504.

[28] **Song, S.**, *Wang, S., Wu, X., Huang, Y., and Fu, B. (2022). "Decreased Virtual Water Outflows from the Yellow River Basin Are Increasingly Critical to China". In: *Hydrology and Earth System Sciences* 26.8, pp. 2035–2044. DOI: 10.5194/hess-26-2035-2022.

[29] Wang, Y., *Liu, Y., **Song, S.**, Yao, Y., and Fu, B. (2022). "A review of community-based social-ecological system adaptation pathways". In: *Progress in Geography* 41.5, pp. 935–944.

[30] Wu, X., *Fu, B., Wang, S., **Song, S.**, Li, Y., Xu, Z., Wei, Y., and Liu, J. (2022). "Decoupling of SDGs Followed by Re-Coupling as Sustainable Development Progresses". In: *Nature Sustainability*. DOI: 10.1038/s41893-022-00868-x.

2021 [31] Gao, D., *Wang, S., Li, Z., Wei, F., Chen, P., **Song, S.**, Wang, Y., Wang, L., and Fu, B. (2021). "Threshold of Vapour–Pressure Deficit Constraint on Light Use Efficiency Varied with Soil Water Content". In: *Ecohydrology*. DOI: 10.1002/eco.2305.

[32] Li, Z., *Wang, S., **Song, S.**, Wang, Y., and Musakwa, W. (2021). "Detecting Land Degradation in Southern Africa Using Time Series Segment and Residual Trend (TSS-RESTREND)". In: *Journal of Arid Environments* 184, p. 104314. DOI: 10.1016/j.jaridenv.2020.104314.

[33] **Song, S.**, *Du, J., Wu, Q., Ni, M., Wang, Y., and Zhang, Y. (2021). "The Responses of *Spinifex Littoreus* to Sand Burial on the Coastal Area of Pingtan Island, Fujian Province, South China". In: *Écoscience*, pp. 1–10. DOI: 10.1080/11956860.2021.1888523.

[34] **Song, S.**, *Wang, S., Fu, B., Dong, Y., Liu, Y., Chen, H., and Wang, Y. (2021). "Improving Representation of Collective Memory in Socio-Hydrological Models and New Insights into Flood Risk Management". In: *Journal of Flood Risk Management* 14.1. DOI: 10.1111/jfr3.12679.

[35] Wang, S., **Song, S.**, Zhang, J., Wu, X., and *Fu, B. (2021). "Achieving a Fit between Social and Ecological Systems in Drylands for Sustainability". In: *Current Opinion in Environmental Sustainability* 48, pp. 53–58. DOI: 10.1016/j.cosust.2020.09.008.

[36] Wang, Y., *Liu, Y., **Song, S.**, and Fu, B. (2021). "Research progress of the water-food-energy-ecosystem nexus". In: *Advances in earth science* 36.07, pp. 684–693.

[37] Yao, Y., Fu, B., *Liu, Y., Wang, Y., and **Song, S.** (2021). "The Contribution of Ecosystem Restoration to Sustainable Development Goals in Asian Drylands: A Literature Review". In: *Land Degradation & Development*, ldr.4065. DOI: 10.1002/ldr.4065.

2020 [38] **Song, S.**, *Wang, S., Fu, B., Liu, Y., Wang, K., Li, Y., and Wang, Y. (2020). "Sediment Transport under Increasing Anthropogenic Stress: Regime Shifts within the Yellow River, China". In: *Ambio* 49.12, pp. 2015–2025. DOI: 10.1007/s13280-020-01350-8.

2019 [39] **Song, S.**, *Wang, S., Fu, B., Chen, H., Liu, Y., and Zhao, W. (2019). "Study on adaptive governance of social-ecological system: Progress and prospect". In: *Acta Geographica Sinica* 74.11, pp. 2401–2410.

[40] Zhang, M., Wang, S., *Fu, B., Wei, X., Wang, C., **Song, S.**, and Wei, F. (2019). "Structure Disentanglement and Effect Analysis of the Arid Riverscape Social-Ecological System Using a Network Approach". In: *Sustainability* 11.19, p. 5159. DOI: 10.3390/su11195159.

2017 [41] Yang, X., *Du, J., Qin, J., Chen, Z., Yang, L., and **Song, S.** (2017). "Diurnal variation characteristics of leaf water potential of Spinifex littoreus on the nebkhas in different succession periods on the coast of Pingtan Island, Fujian Province, China". In: *Chinese Journal of Applied Ecology* 28.10, pp. 3260–3266.

B) Publications without peer review process

2024 [1] **Song, S.**, Wang, S., Jiao, C., and Mantilla, E. J. (2024). "Empowering Human-Water System Analysis through ABSESpy: An Agent-Based Modeling Framework of SES". In: EGU24. Copernicus Meetings. DOI: 10.5194/egusphere-egu24-5635.

2023 [2] **Song, S.**, Wang, S., and Fu, B. (2023). "Institutional Impacts on the Evolution of the Yellow River, China: A Perspective from Socio-Hydrological Modelling". In: EGU2023. Vienna, Austria: Copernicus Meetings. DOI: 10.5194/egusphere-egu23-4221.

2019 [3] **Song, S.**, Wang, S., Fu, B., Yanxu, L., Kevin, W., Yikai, L., and Yaping, W. (2019). "Sediment Transport under Increasing Anthropogenic Stress: Regime Shifts Within the Yellow River, China". In: AGU Fall Meeting 2019. San Francisco, USA: AGU.

*** = Corresponding author**

The following above-mentioned publications have evolved from my doctoral dissertation:

- "Water Quotas Improve Irrigation but Undermine Groundwater Sustainability" (under review)
- "Institutional Impacts on the Evolution of the Yellow River, China" (2023)
- "Identifying Regime Transitions for Water Governance in the Yellow River Basin, China" (2023)
- "Study on adaptive governance of social-ecological system: Progress and prospect" (2019)